

CBSE Class 12 Physics Previous Year Question Paper – Set 55/1/1

Time Allowed: 3 hours

Maximum Marks: 70

General Instructions

- All questions are compulsory.
- Internal choice is provided wherever applicable.
- Questions follow the current CBSE syllabus.

SECTION A – Objective Questions

1. The force between two parallel current carrying wires is used to define which fundamental physical quantity?

- (a) Electric current
- (b) Charge
- (c) Mass
- (d) Length

2. In a uniform electric field, the work done in moving a charge along an equipotential surface is

- (a) Maximum
- (b) Zero
- (c) Minimum
- (d) Infinite

3. Which of the following physical quantities does not change on refraction of light?

- (a) Speed
- (b) Wavelength
- (c) Frequency
- (d) Direction

4. Two identical capacitors in parallel have an equivalent capacitance of $6\ \mu\text{F}$. What will be their equivalent capacitance if connected in series?

- (a) $1.5\ \mu\text{F}$
- (b) $3\ \mu\text{F}$
- (c) $12\ \mu\text{F}$
- (d) $6\ \mu\text{F}$

5. The intensity at the central maximum in a single-slit diffraction pattern is I_0 . What is the intensity at the first minima?

- (a) $I_0/2$
- (b) I_0
- (c) 0
- (d) $I_0/4$

6. When does a transformer work on maximum efficiency?

- (a) Output load is zero
- (b) Output load is maximum
- (c) Iron losses are equal to copper losses
- (d) Iron losses exceed copper losses

7. Which color of visible light has minimum wavelength?

- (a) Red
- (b) Green
- (c) Blue
- (d) Violet

8. Photoelectric effect supports which nature of light?

- (a) Wave
- (b) Corpuscular
- (c) Particle

(d) Longitudinal

9. Which of the following is correct regarding p-type semiconductors?

- (a) Electrons are majority carriers
- (b) Holes are majority carriers
- (c) Electrons are minority carriers
- (d) Both (b) and (c)

10. The time during which the current in an L-R circuit rises from zero to half its maximum value is (where τ is time constant):

- (a) $\tau \ln 2$
- (b) $\tau/2$
- (c) $\tau \ln 0.5$
- (d) $\tau/\ln 2$

11. If the frequency of incident light in a photoelectric experiment is doubled, what happens to the stopping potential?

- (a) Doubles
- (b) Remains same
- (c) Increases
- (d) Decreases

12. Electrostatic force between two charges 1 C each separated by 1 m in vacuum is

- (a) $9 \times 10^9 \text{ N}$
- (b) $9 \times 10^3 \text{ N}$
- (c) $9 \times 10^6 \text{ N}$
- (d) $9 \times 10^{12} \text{ N}$

13. The SI unit of magnetic flux is:

- (a) Weber
- (b) Tesla

- (c) Henry
- (d) Gauss

14. The poles of a magnet cannot be separated because:

- (a) Magnetic monopoles exist
- (b) They are made together
- (c) Magnetic poles always exist in pairs
- (d) Magnets are brittle

15. The direction of induced current in a conductor by a magnetic field is given by:

- (a) Fleming's right hand rule
- (b) Fleming's left hand rule
- (c) Lenz's law
- (d) Ampere's law

16. On which factor does drift velocity of free electrons not depend?

- (a) Mobility
- (b) Number density
- (c) Area of conductor
- (d) Electric field

17. Which among the following is not a property of electromagnetic waves?

- (a) They are transverse
- (b) Travel at constant speed in vacuum
- (c) Require material medium
- (d) Carry energy

18. The energy equivalent of 1 amu is:

- (a) 931 MeV
- (b) 1 eV
- (c) 1 MeV

(d) 1860 MeV

19. Assertion (A): In a good conductor, electric field inside is zero in static condition.

Reason (R): In a good conductor, free electrons rearrange themselves to cancel internal electric field.

- (a) Both A and R are true & R is correct explanation of A
- (b) Both A and R are true but R is not explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

20. Assertion (A): The potential energy of an electron in a hydrogen atom is negative.

Reason (R): Electron is bound to nucleus by electrostatic force.

- (a) Both A and R are true & R is correct explanation of A
- (b) Both A and R are true but R is not explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

SECTION B – Short Answer Questions (2 marks)

21. Define electric flux and write its SI unit.

22. State Lenz's law and explain its significance in electromagnetic induction.

23. Why does a ray of light split into different colors when it passes through a prism?

24. Write one difference between a conductor and a dielectric with respect to polarization.

25. State any two properties of equipotential surfaces.

26. For what condition does wheatstone bridge become balanced? Write its mathematical equation.

- 27. Give two differences between photo-diodes and ordinary diodes.**
- 28. What is the use of a choke coil in an AC circuit? How is it preferred over a resistor?**
- 29. Define magnifying power of a simple microscope. State its expression.**
- 30. State the second law of photoelectric emission and write its mathematical relation.**
- 31. Why can't we use ordinary glass lens for ultraviolet region?**
- 32. Write the expression for the force between two charges in a medium of dielectric constant K and compare it with that in vacuum.**
- 33. What is meant by forward biasing of a p-n junction diode? How does it affect the width of depletion region?**
- 34. How does current in a conductor depend on number density and drift velocity of electrons?**
- 35. State any two differences between conductors and insulators.**
- 36. Write two applications of superconductors in our daily life.**
- 37. How does a zener diode work as a voltage regulator?**
- 38. What is mutual induction? State one application.**
- 39. What will happen to the capacitance of a parallel plate capacitor if area of plates is doubled while distance between them is halved?**

40. Draw the energy band diagram for a semiconductor.

SECTION C – 3-Mark Questions

41. State Gauss's law. Using the law, derive an expression for electric field due to a long straight charged wire.

42. Explain how an ammeter is converted into a voltmeter.

43. Describe the construction and working principle of a cyclotron.

44. With a diagram, explain the process of forward and reverse biasing in a p-n junction.

45. Explain Brewster's law and its significance in polarization of light.

46. State and derive the expression for the resonance condition in a series LCR circuit.

47. Describe how radioactive decay follows an exponential law. Derive the relation.

48. State de Broglie hypothesis and write the expression for de Broglie wavelength.

49. Explain the construction, principle, and working of a moving coil galvanometer.

50. Describe binary fission with the help of a suitable example from the nuclear domain.

SECTION D – Case Study / Source-Based Questions

51. Read the following passage and answer the questions below:

"A student designs a parallel plate capacitor using air as dielectric. She observes that inserting a plastic sheet of thickness equal to the separation between plates

increases the capacitance. She also notices that connecting the capacitor across a battery and then disconnecting, the charge on plates behaves differently with and without the sheet."

- (a) Why does capacitance increase on inserting the plastic sheet?
- (b) What happens to the energy stored if the capacitor remains isolated after inserting the sheet?
- (c) Under what condition will the charge on plates remain constant?
- (d) How would the potential difference change across the plates in this case?

SECTION E – Long Answer Questions (5 marks)

52. Explain the principle, construction, and working of a full wave rectifier. Compare the rectification efficiency with a half wave rectifier and elaborate on the use of filters in smoothing the output. Discuss how the choice of components can impact the reliability of the device in practical circuits.

53. Discuss in detail the phenomenon of interference of light using Young's double slit experiment. How is the fringe width calculated and what factors affect it? Explain how this experiment demonstrates the wave nature of light. Also mention any two applications of optical interference in technology.

Answer Key – CBSE Class 12 Physics – Set 55/1/1

1. (a)
2. (b)
3. (c)
4. (a)
5. (c)
6. (c)
7. (d)
8. (c)
9. (d)
10. (a)
11. (c)
12. (a)
13. (a)
14. (c)
15. (a)
16. (c)
17. (c)
18. (a)
19. (a)
20. (a)

21. Electric flux is the measure of the total electric field passing through a given area. Its SI unit is Nm^2/C .
22. Lenz's law states that induced EMF opposes the change causing it. It ensures conservation of energy in electromagnetic induction.
23. Dispersion causes light to split into different colors due to variation in refractive index for different wavelengths.
24. In conductors, free charges move freely; in dielectrics, atoms polarize but charges do not move freely.
25. Equipotential surfaces never intersect and electric field is always perpendicular to them.
26. Wheatstone bridge is balanced when the ratio of resistances is equal; $(R_1/R_2) = (R_3/R_4)$.
27. A photodiode is sensitive to light and generates current, ordinary diode does not. Photodiode operates in reverse bias.
28. A choke coil controls current with negligible power loss; it is preferred over resistor as it doesn't dissipate energy as heat.
29. Magnifying power is the ratio of angular size of image to object; $m = 1 + (D/f)$.
30. The second law states that photoelectric current is proportional to intensity; $I \propto \text{Intensity}$.
31. Ordinary glass absorbs ultraviolet, so it can't transmit UV rays.
32. $F = (1/4\pi\epsilon_0 K)(q_1 q_2 / r^2)$; force in medium is reduced by factor K compared to vacuum.
33. Forward bias is connecting p to +, n to -; it reduces depletion width.
34. Current $I = neAv_d$, where n is number density, e is charge, A is area, v_d is drift velocity.

35. Conductors have free electrons, insulators do not; conductors allow current, insulators resist current.
36. Used in MRI machines and Maglev trains.
37. Zener diode maintains constant output voltage by breakdown in reverse bias.
38. Mutual induction is generation of EMF in one coil due to changing current in another; used in transformers.
39. Capacitance increases by a factor of 4.
40. The valence band is filled, conduction band is partly filled or empty, with small energy gap in between.
41. Gauss's law: Electric flux through closed surface equals charge/ ϵ_0 ; for long wire, $E = \lambda / (2\pi\epsilon_0 r)$.
42. An ammeter is converted into a voltmeter by connecting a large resistance in series; it then measures potential difference.
43. Cyclotron accelerates charged particles using perpendicular electric and magnetic fields; particles follow spiral path to gain energy.
44. In forward bias, p is connected to positive, depletion region narrows; in reverse bias, connected to negative, depletion widens.
45. Brewster's law states reflected and refracted rays are perpendicular at polarizing angle; light reflected is plane polarized.
46. Resonance in LCR circuit occurs when $X_L = X_C$; total impedance minimum and current maximum.
47. Radioactive decay rate decreases exponentially: $N = N_0 e^{-\lambda t}$; λ is decay constant.
48. De Broglie hypothesis: matter has wave nature; $\lambda = h/p$, where h is Planck's constant.

49. Moving coil galvanometer has a coil in magnetic field; current causes torque, coil turns, deflection proportional to current.

50. Binary fission in nuclear domain is seen in uranium-235 splitting into smaller nuclei on neutron absorption.

51. (a) Capacitance increases because plastic is a dielectric with higher constant than air.

(b) Energy stored decreases as capacitance increases while charge stays constant.

(c) If capacitor is isolated (battery disconnected), charge remains constant.

(d) Potential difference decreases with increased capacitance.

52. A full wave rectifier uses both halves of AC, with two diodes in bridge or center-tapped configuration. It provides higher rectification efficiency (~81%) than half wave. Filters (like capacitor filters) smooth output, reducing ripples. Good thermal rating and reliable diodes ensure device stability.

53. Young's double slit experiment shows interference by producing alternate bright and dark fringes. Fringe width $\beta = \lambda D/d$ where λ is wavelength, D distance to screen, d is slit separation. Higher λ or D or smaller d increases β . Demonstrates wave nature of light. Applications include interference filters and anti-reflective coatings.